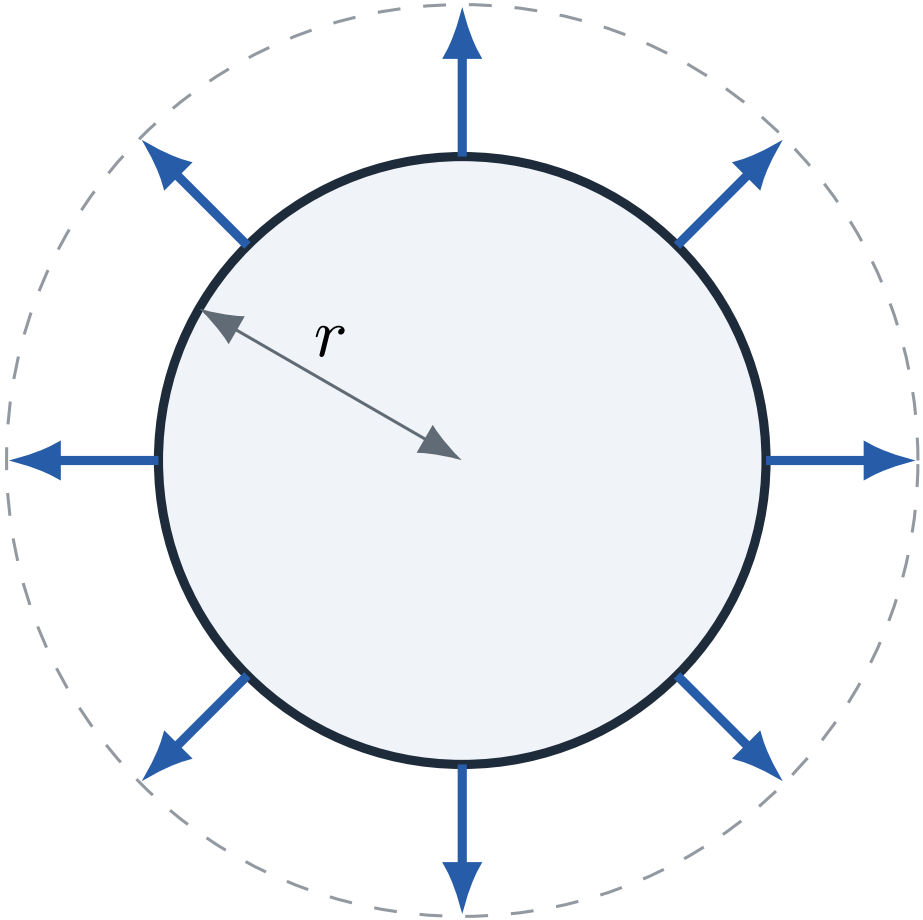

$$u = \beta T^4, \quad P = u/3$$

$$U = uV, \quad \dot{r} = \alpha r$$

绝热准静态膨胀